

System Analysis & Design

BCA — Semester III — 2022 — Answer Sheet

Subject Code: CACS203

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. You have been hired as a project manager of a large government project to replace an aging air traffic control system. Would you choose waterfall or agile as your application development method? Explain your decision.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the roles and responsibilities of software project manager in project development.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Giving a simple example, explain how ER schemas are converted into a set of relations. Your example should have the use of multivalued and composite attributes too. Cardinality is a must.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is meant by deliverable? Explain the major deliverables of planning phase.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is the significance of normalization? Explain 2NF giving a suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain different approaches to installation? Which one is most expensive and risky?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. A system costs RS 2,00,000 to install. The profit per year is RS. 50,000. Assuming an interest rate is 6%, what is the payback period of the investment?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. List rules for diagramming DFDs. Draw a context diagram and expand it to top level logical DFD for Student Enrollment System. The enrollment process works as follows.

- Students send in an application form containing their personal details and their desired course.
- The university checks that the course is available and that the student has necessary academic qualifications.
- If the course is available, the student is enrolled in the course, and the university confirms the enrollment by sending a confirmation letter that they are registered for the course.
- If the course is unavailable, the student is sent a rejection letter.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain the factors that influence the cost of maintenance. Explain the steps involved in maintenance process and contrast them with the phases of system development lifecycle.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What qualities a software tester should have? What are the deliverables from coding, testing and Installation? Explain different types of Software testing in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.